DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. ETC **Semester: VIII**

Subject Code & Name: Biomedical Signal Processing (BTETPE801C)

Max Marks: 60 **Date:29/07/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All Questions are Compulsory*
2. *Draw neat diagram wherever necessary.*
3. *Figures to right indicates full marks*
4. *Assume suitable data wherever necessary and mention it clearly*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	What is Pan-Tompkin algorithm.		06
B)	Draw Electrocardiogram waveform and write Amplitude, duration of each interval.		06
C)	Explain "Homomorphic filtering"		06
Q.2	**Solve Any Two of the following.**		
A)	Explain CAD based on Biomedical Signal in detail with block diagram.		06
B)	What is Blackman-Turkey Transformation?		06
C)	Explain Nervous System in the Human body.		06
Q. 3	**Solve Any Two of the following.**		
A)	Explain Least Mean Square (LMS) adaptive filtering.		06
B)	Define Periodogram and write application of it.		06
C)	Explain Cardiovascular System and Electrocardiogram		06
Q.4	**Solve Any Two of the following.**		
A)	Explain 10-20 electrode placement system of EEG with neat diagram.		06
B)	Explain the CAD based on Biomedical Signal with block diagram.		06
C)	What are the different types of Windowing techniques used in Biomedical Signal.		06
Q. 5	**Solve Any Two of the following.**		
A)	With neat diagram explain the "How Action Potential (PQRST waveform) generated and cardiac system works".		06
B)	Define the following terms: ECG, EEG, EMG, EOG, ENG, ERP		06
C)	Explain the following terms: Bradycardia, Tachycardia, Sinus Arrhythmia		06

***** End *****